

AI OPERATIONS ASSISTANT

SaaS Platform — Product Requirements Document

v1.0

2026

MVP SCOPE

"An AI employee that helps small businesses handle repetitive work — intelligently, automatically, and at scale."

90-Day Build

6 Core Modules

SMB-Focused

Full-Stack SaaS

1. Project Overview

The AI Operations Assistant is a multi-module SaaS platform designed to replace repetitive manual tasks in small and medium businesses (SMBs). It combines document intelligence, automated customer support, lead processing, and admin automation — all accessible through a clean web dashboard. The platform is built on a Retrieval-Augmented Generation (RAG) engine and is extensible via n8n workflow automation.

Problem Statement

Small businesses in Bangladesh and globally struggle with repetitive operational tasks: answering the same customer questions, manually processing leads, onboarding new staff, and generating routine documents. Hiring staff for these tasks is expensive; most SMBs lack the budget or technical team to deploy AI solutions themselves. This platform solves that gap.

Solution

Without This Platform	With This Platform
Staff manually answers repeat FAQs	AI instantly answers from uploaded docs
Leads manually reviewed and summarized	AI extracts key info automatically
New staff takes weeks to onboard	Staff queries the knowledge base in seconds
Reports manually compiled	AI generates summaries on demand
Email replies typed from scratch	AI drafts professional replies instantly

2. Target Users

Segment	Primary Use Case
Agencies & Consulting Firms	Need automated client reporting, lead handling, and knowledge bases for multiple clients.
Coaching & Training Centers	Require FAQ bots, student onboarding docs, and schedule assistants.
Clinics & Healthcare Offices	Use document search for procedures, patient FAQs, and appointment-related email drafts.
E-commerce Stores	Need lead qualification, order-related FAQ handling, and policy-based customer support.
Law Firms & Real Estate Offices	Heavy document workflows — case summaries, contract Q&A, client intake automation.

3. Core Features (MVP)

Module 1 Document Intelligence (RAG Engine)

- Users upload company documents: PDFs, Word docs, policy files, SOPs, manuals.
- Files are chunked, embedded, and stored in a FAISS or Chroma vector store.
- Users can ask natural-language questions; the AI retrieves relevant chunks and answers with source citations.
- Supports multi-document workspaces per business account.
- Admin can tag, delete, and update documents through the dashboard.

Module 2 Customer Support AI

- Embeds FAQ knowledge into a chat interface that businesses can use internally.
- Answers customer queries using only the company's uploaded documents.
- Falls back gracefully when the answer is not in the knowledge base.
- Logs all unanswered queries so the business can fill knowledge gaps.
- Optionally integrates with n8n to push queries from email/WhatsApp into the assistant.

Module 3 Email Draft Generator

- User pastes a raw customer email or inquiry into the interface.
- AI generates a professional reply based on company tone and uploaded context.
- User can edit, regenerate, or copy the draft with one click.
- Optional tone selector: Formal, Friendly, Concise.
- Can be automated via n8n: incoming Gmail → AI draft → saved as draft in Gmail.

Module 4 Lead Summarizer & Qualifier

- User pastes a raw lead inquiry or form submission.
- AI extracts: name, interest, budget, urgency, and key questions.
- Outputs a structured lead card: summary, recommended action, priority score.
- Leads are stored in a leads database within the dashboard.
- Optionally automated via n8n: form submission → AI summary → CRM entry.

Module 5 Internal Knowledge Base

- Acts as a smart onboarding assistant for new staff.
- Staff can ask: 'What is our refund policy?' or 'How do I handle a complaint?'
- AI answers from company-uploaded docs with exact source references.
- Reduces onboarding time significantly for growing teams.
- Admins can version-control documents (update old → archive old chunks).

Module 6 Admin Report & Summary Generator

- User pastes meeting notes, raw data, or a transcript.
- AI generates a clean structured summary with action items and highlights.
- Can produce: weekly summaries, meeting minutes, client reports.
- Optionally triggered via n8n on a schedule (daily/weekly digest).

- Output can be copied, downloaded, or sent via email through the dashboard.

4. Technology Stack

Layer	Technology	Purpose
Frontend	React + Tailwind CSS	User dashboard, chat UI, forms
Backend	FastAPI (Python)	REST API, business logic, auth
Database	PostgreSQL	Users, leads, docs metadata, logs
Vector Store	FAISS / Chroma	Embedding storage for RAG
AI Layer	OpenAI GPT-4o API	LLM completions, embeddings
Embeddings	text-embedding-3-small	Document vectorization
Automation	n8n (self-hosted or cloud)	Email, lead, and scheduling workflows
Auth	JWT + bcrypt	User login, session management
Hosting (FE)	Vercel	Frontend deployment
Hosting (BE)	Render / Railway	Backend API deployment
File Storage	AWS S3 / Cloudflare R2	Document upload storage

n8n Integration Strategy: n8n handles all automation triggers and background workflows — it is the automation layer, not the app layer. The React + FastAPI stack handles the user-facing product, billing, auth, and analytics. n8n connects external tools (Gmail, WhatsApp, Google Sheets, CRMs) to your FastAPI backend via webhooks. This hybrid approach allows rapid feature delivery without sacrificing engineering depth.

5. Database Schema (Core Tables)

Table: users

Column	Type	Description
id	UUID	Primary key
email	VARCHAR(255)	Unique login email
password_hash	TEXT	bcrypt hashed password
business_name	VARCHAR(255)	Business display name
plan	VARCHAR(50)	free / pro / enterprise
created_at	TIMESTAMP	Account creation time

Table: documents

Column	Type	Description
id	UUID	Primary key
user_id	UUID FK	Owner reference
filename	TEXT	Original file name
storage_url	TEXT	S3/R2 file URL
chunk_count	INTEGER	Number of embedded chunks
status	VARCHAR(50)	processing / ready / error
uploaded_at	TIMESTAMP	Upload timestamp

Table: chat_logs

Column	Type	Description
id	UUID	Primary key
user_id	UUID FK	Owner reference
question	TEXT	User's query
answer	TEXT	AI-generated response
source_doc_ids	TEXT[]	Docs used in answer
module	VARCHAR(50)	doc_qa / support / etc.
created_at	TIMESTAMP	Query timestamp

Table: leads

Column	Type	Description
id	UUID	Primary key
user_id	UUID FK	Owner reference

raw_input	TEXT	Original inquiry text
name	TEXT	Extracted lead name
interest	TEXT	Extracted product/service interest
budget	TEXT	Extracted budget mention
urgency	VARCHAR(50)	high / medium / low
priority_score	INTEGER	AI-generated 1–10 score
created_at	TIMESTAMP	Submission timestamp

6. Key API Endpoints

Method	Endpoint	Description
POST	/auth/register	Register new user account
POST	/auth/login	Login, returns JWT token
POST	/documents/upload	Upload document file
GET	/documents/	List all user documents
DELETE	/documents/{id}	Delete document and embeddings
POST	/chat/query	RAG-based document Q&A
POST	/support/answer	Customer support AI response
POST	/email/draft	Generate email reply draft
POST	/leads/summarize	Extract and score a lead
GET	/leads/	List all stored leads
POST	/reports/summarize	Generate meeting/report summary
GET	/dashboard/stats	Usage stats for dashboard

7. UI Pages & Dashboard Layout

Login / Register Page

- Email + password fields with JWT-based auth.
- Link to register new business account.
- Redirect to Dashboard on success.

Dashboard (Home)

- Welcome card with business name.
- Quick stats: docs uploaded, queries made, leads captured.
- Shortcut buttons to all 6 modules.
- Recent activity feed.

Document Manager

- Upload button (drag-and-drop) supporting PDF, DOCX, TXT.
- List view of all uploaded docs with status badge.
- Delete and preview options per document.
- Processing indicator for newly uploaded files.

AI Chat (Doc Q&A;)

- Chat-style interface (user bubble + AI bubble).
- Source citation shown below each AI answer.
- Conversation history within session.
- Clear chat and new session buttons.

Email Draft Generator

- Text area: 'Paste customer email here'.
- Tone selector: Formal / Friendly / Concise.
- Generate button → displays draft below.
- Copy to clipboard and Regenerate buttons.

Lead Summarizer

- Paste raw inquiry text.
- Click Analyze → AI returns structured lead card.
- Lead card shows: Name, Interest, Budget, Urgency, Priority Score.
- Save to Leads Database button.

Leads Dashboard

- Table of all saved leads with filter + sort.
- Click any lead to expand full summary.
- Export to CSV option.

Report / Summary Generator

- Paste raw notes, transcript, or meeting text.
- Click Summarize → AI returns structured summary + action items.
- Copy and Download as TXT options.

Settings

- Business profile: name, logo, tone preference.
- API usage stats and plan info.
- Delete account option.

8. 90-Day Build Roadmap

Week	Focus	Deliverables	Status
1	Project Setup	Repo init, FastAPI skeleton, React app, DB setup, auth endpoints	Todo
2	Document Upload	File upload API, S3 integration, chunking pipeline, embedding storage	Todo
3	RAG Engine	FAISS/Chroma integration, semantic search, Q&A endpoint + basic UI	Todo
4	Chat UI	React chat interface, source display, conversation logging	Todo
5	Email Draft	Email generator endpoint, tone selector UI, copy/regenerate features	Todo
6	Lead Summarizer	Lead extraction endpoint, lead card UI, leads database + list view	Todo
7	Report Generator	Summary endpoint, paste interface, download feature	Todo
8	n8n Workflows	Email → AI draft workflow, form → lead workflow, webhook endpoints	Todo
9	Auth + Dashboard	JWT auth, full dashboard home, stats API, settings page	Todo
10	Testing	Unit tests, integration tests, bug fixes, mobile responsiveness	Todo
11	Deploy	Vercel deploy (FE), Render deploy (BE), domain, SSL, env config	Todo
12	Launch Prep	Landing page, demo video, README, GitHub publish, first outreach	Todo

Month 1: Core Foundation

Month 2: Features + Automation

Month 3: Polish + Launch

9. Resume & Portfolio Value

How to Describe This Project

Job Title Target: AI Engineer / Full-Stack Developer / ML Engineer

Bullet 1: Built a full-stack AI Operations SaaS for SMBs featuring RAG-based document intelligence, automated customer support, and lead processing — deployed on Vercel + Render with FastAPI and React.

Bullet 2: Designed and implemented a multi-module AI platform integrating OpenAI GPT-4o, FAISS vector search, PostgreSQL, and n8n automation workflows for real business use cases.

Bullet 3: Architected end-to-end RAG pipeline with document chunking, embedding generation, semantic retrieval, and cited AI responses — used by small businesses to automate 40%+ of repetitive operational tasks.

Skills Demonstrated

Skill Area	Evidence from This Project
Backend Engineering	FastAPI REST API with auth, file handling, and AI endpoints
Frontend Development	React dashboard with chat UI, forms, and real-time updates
AI / LLM Integration	RAG pipeline with OpenAI embeddings and GPT-4o completions
Database Design	PostgreSQL schema with relational data, UUID keys, and indexing
DevOps / Deployment	Cloud deployment, environment config, CI/CD basics on Render + Vercel
Automation Engineering	n8n workflows connecting external tools to your backend
Product Thinking	End-to-end SaaS with real user flows, not just a script

10. Monetization Strategy

Plan	Price	Limits	Target
Free	■0 / mo	3 docs, 50 queries/mo, 1 user	Testing / Trial
Starter	■999 / mo (~\$9)	20 docs, 500 queries/mo, 3 users	Solo operators
Pro	■2,499 / mo (~\$23)	100 docs, 2,000 queries/mo, 10 users	SMBs, agencies
Enterprise	Custom	Unlimited + white-label + API access	Larger clients

Honest Risk Assessment

■ Over-engineering risk	Start with Module 1 (Doc Q&A;) only. Ship it. Then add modules one at a time.
■ API cost risk	Set per-user query limits. Use caching for repeated queries. Monitor OpenAI spend weekly.
■ Competitor risk	You're not competing with ChatGPT. You're serving SMBs who want a ready-made tool for their business, not a general AI.
■ Scope creep	Use this PRD as your contract with yourself. Do not add features outside this document until v1 is live.

11. Week-by-Week Execution Checklist

Week 1 — Project Foundation

- Initialize GitHub repo with monorepo structure (frontend/, backend/)
- Create FastAPI project skeleton with folder structure
- Set up PostgreSQL locally and on cloud (Railway or Supabase)
- Create users table + JWT auth endpoints (/register, /login)
- Initialize React app with Tailwind CSS
- Build Login and Register pages in React
- Connect React to FastAPI auth endpoints
- Test auth flow end to end

Week 2 — Document Upload Pipeline

- Add documents table to PostgreSQL
- Build /documents/upload endpoint in FastAPI (accept PDF, DOCX, TXT)
- Integrate S3 or Cloudflare R2 for file storage
- Build document chunking logic (split by paragraph/token count)
- Generate embeddings using OpenAI text-embedding-3-small
- Store chunks + embeddings in FAISS or Chroma
- Build Document Manager UI in React (upload + list view)
- Add delete document functionality

Week 3–4 — RAG Q&A; Engine + Chat UI

- Build /chat/query endpoint: receive question → embed → retrieve → GPT answer
- Include source citation in API response
- Store all queries in chat_logs table
- Build React Chat UI: message bubbles, source display
- Test with multiple document types and queries
- Handle edge case: question not in any document

Week 5–6 — Email Draft + Lead Summarizer

- Build /email/draft endpoint with tone parameter
- Prompt engineer the draft for business-appropriate output
- Build Email Draft UI: paste area, tone selector, draft display
- Build /leads/summarize endpoint: extract name, interest, budget, urgency
- Generate priority score (1–10) with brief justification
- Store lead to leads table
- Build Lead Card UI and Leads Dashboard list view

Week 7–8 — Report Generator + n8n Integration

- Build /reports/summarize endpoint: structure raw notes into summary + action items
- Build Report Generator UI: paste area + output display
- Set up n8n instance (cloud or self-hosted)
- Create n8n workflow: Gmail trigger → /email/draft → save draft

- Create n8n workflow: form submission → /leads/summarize → CRM or Notion
- Test both n8n workflows end to end
- Document webhook endpoints for n8n connections

Week 9–10 — Dashboard + Testing

- Build Dashboard home with stats: queries, leads, docs
- Build Settings page: business name, tone preference
- Add API usage tracking to backend
- Write unit tests for all endpoints
- Fix bugs found during testing
- Ensure mobile responsiveness on all pages

Week 11–12 — Deploy + Launch

- Deploy frontend to Vercel with custom domain
- Deploy backend to Render with environment variables set
- Configure CORS, SSL, production PostgreSQL
- Record a 2-minute demo video of all features
- Write a clean README.md with setup instructions
- Publish GitHub repo (public)
- Write a LinkedIn post or landing page about the product
- Reach out to 3–5 potential users for feedback

Final Note

You are not building a side project. You are building a product.

Every week you ship a working feature, you have something real to show — to employers, to customers, and to yourself. The goal is not perfection. The goal is a live, working SaaS product that solves a real problem for real businesses.

Use this document as your source of truth. When in doubt, refer back here. When you feel lost, open Week 1 and do the next unchecked task.

Build it. Ship it. Own it.